TEAM LEARNING PROJECT — NO. 3



Tarmo

A simplified team task board — built with Python (Flask) and MongoDB to practice full-stack development, Git collaboration, and production-grade coding standards.

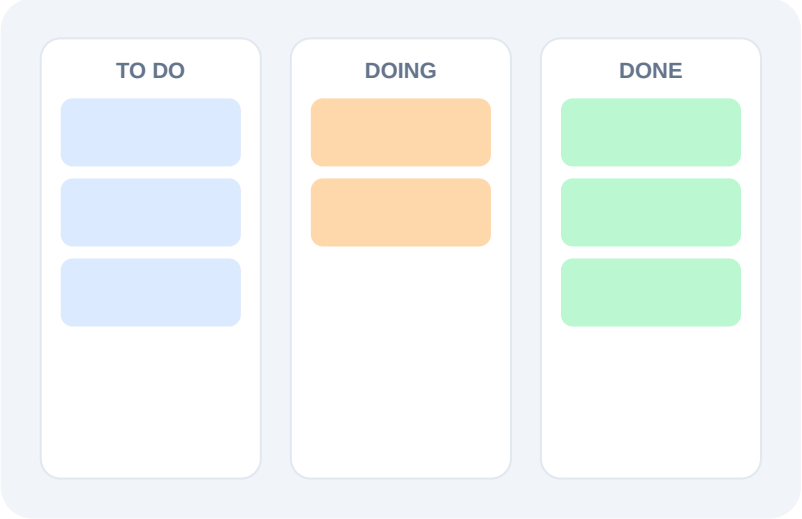
Version 1 · Limited Feature Set · 4-Person Team

OVERVIEW

1 What Is This Project?

A lightweight clone of Trello — a visual task board where work is organized into **boards**, each containing **lists** (columns like "To Do", "Doing", "Done"), which hold **cards** representing individual tasks.

The goal isn't to replace Trello — it's a structured way for a 4-person team to practice real-world skills: a Flask backend, a real cloud database, team-based frontend/backend collaboration, and secure coding standards used in production applications.



- Used as a hands-on learning exercise, not a commercial product
- Version 1 intentionally limits scope to stay achievable
- Each teammate owns an independent part of the system

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TECHNOLOGY

What It's Built With

Py

Python + Flask

The backend web framework. Handles all logic: creating boards, moving cards, auth checks.

DB

MongoDB (Atlas)

Cloud NoSQL database. Stores users, boards, lists, and cards as JSON-like documents.

UI

HTML + Tailwind CSS

The visual interface — pages rendered server-side with Jinja templates.

JS

JavaScript + SortableJS

Enables drag-and-drop of cards between lists directly in the browser.

🔒

Werkzeug Security

Hashes every password before storage — plain-text passwords are never saved.

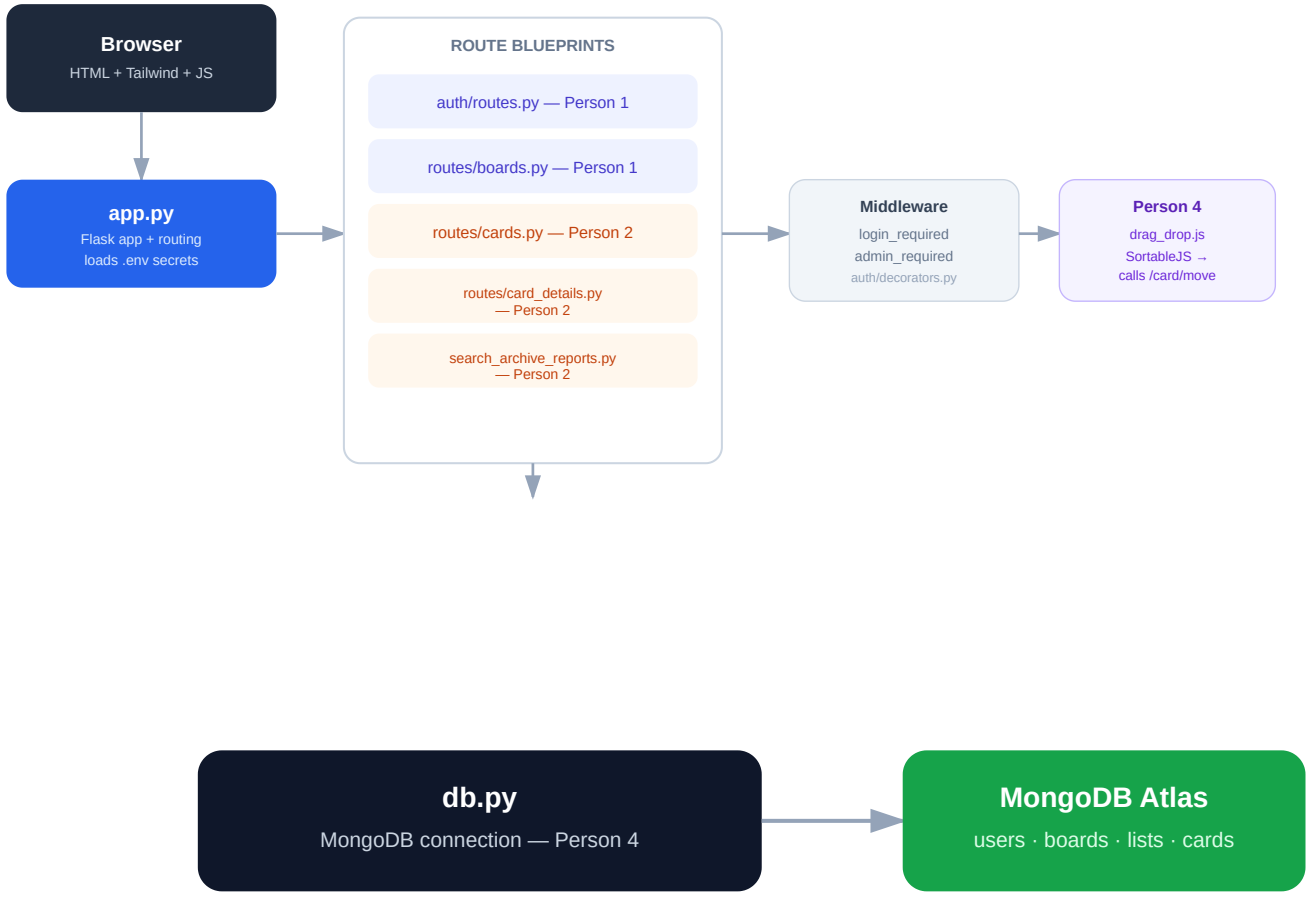
.env

python-dotenv

Keeps secrets (DB credentials, secret key) out of source code and off GitHub.

SYSTEM DESIGN

3 How the Pieces Connect



Every route reads its connection string from `.env` — never hardcoded — via `db.py`

SCOPE

4 Version 1 Features

- 👤 Register / Login
- 🛡️ Admin Role
- 📋 Multiple Boards
- 📑 Lists (Columns)
- 🗂️ Create / Delete Cards
- 🖱️ Drag & Drop Cards
- 🏷️ Color Labels
- 📅 Due Dates
- 🙋 Assign Members
- 💬 Comments
- 🔍 Search Cards
- 🗄️ Archive Cards
- 📊 Board Stats
- 🔐 Hashed Passwords
- 🌱 .env Secrets
- 🚧 Route Middleware

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COLLABORATION

Team & Responsibilities

🐍

Person 1

Backend — Python/Flask

- Register / Login / Logout
- Password hashing
- Boards & Lists
- Multi-board dashboard

🐍

Person 2

Backend — Python/Flask

- Cards (create/move/edit/delete)
- Labels, due dates, assignment
- Comments
- Search, archive, stats

🎨

Person 3

Frontend — HTML/CSS

- All page templates (Jinja)
- Tailwind-based design
- Login/register/dashboard UI
- Board & card layout

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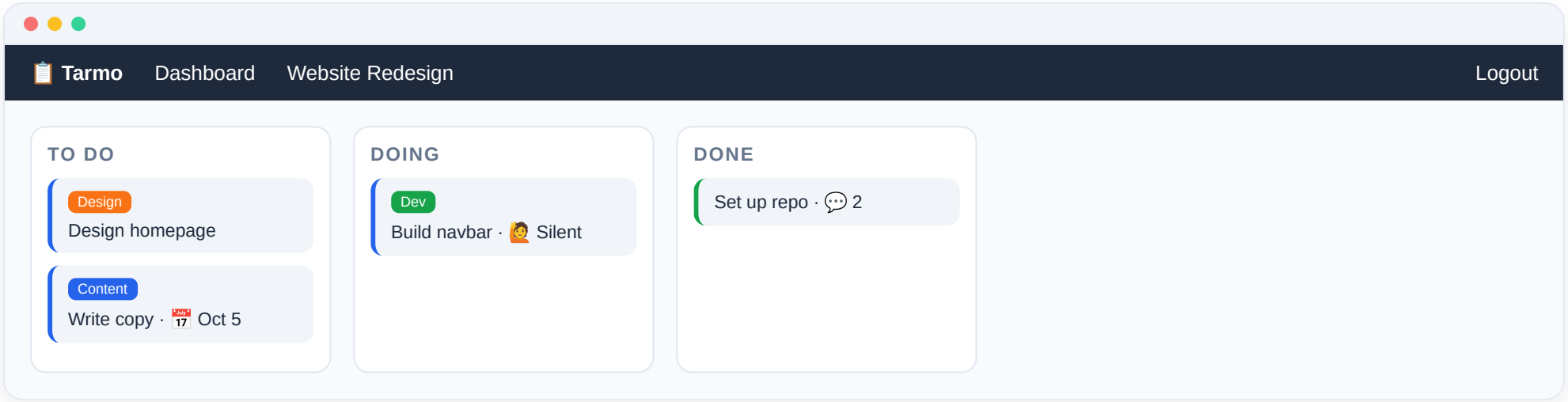
Person 4

Database + JavaScript

- MongoDB Atlas setup
- Schema design (SCHEMA.md)
- db.py connection module
- Drag-and-drop (SortableJS)

RESULT

6 What the Output Looks Like



Wireframe preview — final look depends on Person 3's Tailwind design

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PROJECT MANAGEMENT

Timeline & Milestones

WEEK 1

Setup & Warm-Up

- Everyone completes their role's warm-up notebook
- Person 4 sets up MongoDB Atlas cluster
- Person 3 starts base templates + Tailwind
- Repo cloned, .env configured by each teammate

✅ **Milestone: App runs locally for everyone**

WEEK 2

Core Backend

- Person 1: auth, boards, lists working
- Person 2: cards create/move/delete working
- Person 3: dashboard + board page styled
- Person 4: schema finalized in SCHEMA.md

✅ **Milestone: Create board → add list → add card works end-to-end**

WEEK 3

Details & Interactivity

- Person 2: labels, due dates, comments, search
- Person 4: drag-and-drop wired to /card/move
- Person 3: polish styling across all pages
- Cross-review each other's Pull Requests

✅ **Milestone: Drag a card between lists and it saves**

WEEK 4

Test & Finish

- Run the full Testing Plan (page 10)
- Fix bugs found during testing
- Final PR reviews and merge to main
- Team demo — everyone shows their part

✅ **Milestone: Full flow works with zero crashes**

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PROJECT MANAGEMENT

Task Board (Trello / GitHub Projects)

Every task below is tracked as its own card on a shared board — moved from **To Do** → **In Progress** → **Review (PR)** → **Done** as work happens.

📋 TO DO

- Set due date on a card — Person 2
- Archive card endpoint — Person 2
- Style login/register pages — Person 3
- Board stats page — Person 2

🔨 IN PROGRESS

- MongoDB Atlas cluster setup — Person 4
- Login / register routes — Person 1
- Create / move card routes — Person 2

👀 REVIEW (PR)

- Board + list templates — Person 3
- login_required decorator — Person 1

✅ DONE

- Project structure + app.py — Shared
- SCHEMA.md draft — Person 4
- .env / secrets setup — Shared

Snapshot example — the real board updates daily as work progresses

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PROJECT MANAGEMENT

Risks & Blockers

RISK	SEVERITY	IMPACT	MITIGATION
Person 4 delays MongoDB Atlas setup	High	Person 1 & 2 can't test against a real database	Use the mock dictionary store from the warm-up notebook until Atlas is ready
Merge conflicts in shared files (app.py, base.html)	Medium	Lost work, wasted time resolving conflicts	Pull latest main before starting any task; keep shared-file edits small
Drag-and-drop (JS) takes longer than expected	Medium	Version 1 demo missing a key interactive feature	Fallback: move cards via a simple dropdown/button instead of drag-and-drop
A teammate falls behind on Flask/Mongo basics	Medium	Their tasks block the rest of the team	Pair up for an hour; revisit the relevant warm-up notebook section together
.env accidentally committed to Git	High	Database credentials exposed publicly on GitHub	Confirm .gitignore before first push; rotate the Atlas password immediately if leaked
Scope creep — team wants to add features mid-project	Low	Version 1 deadline slips	New ideas go on a "Version 2" list; Version 1 scope stays frozen

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PROJECT MANAGEMENT

Communication Plan

Weekly Check-In

Once a week, everyone posts in the group: what they finished, what they're working on next, and anything blocking them.

Pull Request Reviews

The Project Manager (or a teammate) reviews every PR before merging — checks it runs, matches the task's TODO steps.

Blocker Escalation

If stuck for more than a day, say so immediately in the group rather than staying quiet — the PM helps unblock or reassigns.

End-of-Week Demo

Each Friday, whoever wants to shows what they built that week — keeps momentum and catches integration issues early.

PROJECT MANAGEMENT

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Testing Plan (Before Calling It Done)

- [ ] **Register** two different users and confirm both can log in
- [ ] Confirm a **wrong password** is correctly rejected
- [ ] Each user **creates a board** and sees only their own boards on the dashboard
- [ ] Create **3 lists** on a board (To Do / Doing / Done)
- [ ] Create several **cards** and move them between lists
- [ ] Add a **label, due date, assignee, and comment** to a card
- [ ] **Search** for a card by keyword and confirm the right one appears
- [ ] **Archive** a card and confirm it disappears from the board but appears in the archive view
- [ ] Confirm a **non-admin** cannot delete a board (middleware check)
- [ ] Confirm **.env** is not present in `git status` / GitHub repo

PROJECT MANAGEMENT

12 Definition of Done & Success Criteria

Tarmo Version 1 is considered **complete and successful** when:

01

All 10 checklist items on the Testing Plan page pass without errors

02

Each of the 4 teammates has completed at least one full Git cycle: clone → branch → commit → push → PR → merge → pull

03

No plain-text passwords or secrets appear anywhere in the GitHub repo

04

The app runs with zero crashes during a full live demo of the core flow

05

Every teammate can explain — in their own words — what their code does and why

06

The team has a working Version 2 wish-list ready, informed by what Version 1 taught them